Your Code

```java
import java.util.*;

class Main {
    static ArrayList<Patient> patientList = new ArrayList<>();
    static HashMap<String, Patient> patientMap = new HashMap<>();

    public static String readNext(Scanner sc) {
        // Reads next token OR next line if tokenizing breaks names
        // Here we want full lines for names/doctors, so we read li
        while (sc.hasNextLine()) {
            String s = sc.nextLine();
            if (s == null) return null;
            s = s.trim();
            if (!s.isEmpty()) return s;
        }
        return null;
    }

    public static String properCase(String name) {
        name = name.trim().replaceAll("\\s+", " ");
        if (name.isEmpty()) return "";
        name = name.toLowerCase();
        String[] words = name.split(" ");
        StringBuilder sb = new StringBuilder();
        for (String w : words) {
            if (w.isEmpty()) continue;
```

Input

```
1
P101
Rahul Kumar
9876543210
```

Output

```
Patient Registered Successfully.
Patient Registered Successfully.
--------------------------------
-----------------------------
ID Name Mobile Department Doctor
```

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```java
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        while (true) {
            String choiceStr = readNext(sc);
            if (choiceStr == null) break;

            int choice;
            try {
                choice = Integer.parseInt(choiceStr);
            } catch (Exception e) {
                System.out.println("Invalid Choice");
                continue;
            }

            switch (choice) {
                case 1: {
                    if (patientList.size() >= 100) {
                        System.out.println("Maximum Patient Limit R
                        break;
                    }

                    String id = readNext(sc);
                    if (id == null || id.trim().isEmpty()) {
                        System.out.println("Invalid Patient ID");
                        break;
                    }
                    id = id.trim();
```

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```
        }
        sc.close();
    }
}

class Patient {
    private String patientId;
    private String patientName;
    private String mobileNumber;
    private String department;
    private String doctorName;

    public Patient(String patientId, String patientName, String mob
                   String department, String doctorName) {
        this.patientId = patientId;
        this.patientName = patientName;
        this.mobileNumber = mobileNumber;
        this.department = department;
        this.doctorName = doctorName;
    }

    public String getPatientId() { return patientId; }
    public String getPatientName() { return patientName; }
    public String getMobileNumber() { return mobileNumber; }
    public String getDepartment() { return department; }
    public String getDoctorName() { return doctorName; }
}
```

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Your Code

```
import java.util.*;

class Main {

    static class Book {
        String bookId;
        String title;
        String author;
        String category;
        String publisher;

        Book(String bookId, String title, String author, String cat
            this.bookId = bookId;
            this.title = title;
            this.author = author;
            this.category = category;
            this.publisher = publisher;
        }
    }

    // Read non-empty line (drives menu and fields safely)
    static String readInput(Scanner sc) {
        while (sc.hasNextLine()) {
            String s = sc.nextLine();
            if (s == null) return "";
            s = s.trim();
            if (!s.isEmpty()) return s;
```

Input

```
1
B101
Java Programming
Herbert Schildt
```

Output

```
Book Added Successfully.
Book Added Successfully.
--------------------------------
--------------------------------
ID TITLE AUTHOR CATEGORY
```

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```java
    static String toTitleCase(String s) {
        s = s.trim().replaceAll("\\s+", " ");
        if (s.isEmpty()) return "";

        s = s.toLowerCase();
        String[] parts = s.split(" ");

        StringBuilder sb = new StringBuilder();
        for (int i = 0; i < parts.length; i++) {
            String w = parts[i];
            if (w.isEmpty()) continue;
            sb.append(Character.toUpperCase(w.charAt(0)));
            if (w.length() > 1) sb.append(w.substring(1));
            if (i < parts.length - 1) sb.append(" ");
        }
        return sb.toString().trim();
    }

    static String normalizeSpaces(String s) {
        return s.trim().replaceAll("\\s+", " ");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        ArrayList<Book> books = new ArrayList<>();
        HashMap<String, Book> bookMap = new HashMap<>(); // Book ID

        // Menu loop
```

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```java
            for (Book b : books) {
                String title = b.title;
                int length = title.length();
                int words = title.isEmpty() ? 0 : title.trim().split("\

                System.out.println("Book Details");
                System.out.println("Book ID : " + b.bookId);
                System.out.println("Title : " + b.title);
                System.out.println("Author : " + b.author);
                System.out.println("Category : " + b.category);
                System.out.println("Publisher : " + b.publisher);

                // extra required string outputs
                System.out.println("Length : " + length);
                System.out.println("Words : " + words);
                System.out.println();
            }
            break;
        }

        case 6: { // Exit
            sc.close();
            return;
        }

        default:
            System.out.println("Invalid Choice");
}

```